

ROHITH APPALA

Backend Developer · Node.js | TypeScript | AWS | GCP | AI/ML Pipelines | RAG & Vector Search

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PROFESSIONAL SUMMARY

Backend Developer with **2+ years of experience** building scalable, serverless and cloud-native applications using **Node.js, TypeScript, AWS, and GCP**. Hands-on expertise in microservices, real-time systems, AI/ML batch inference pipelines, Retrieval-Augmented Generation (RAG) pipelines, and vector search (**Pinecone, ChromaDB**). Designed and deployed large-scale job description processing pipelines using **AWS Bedrock and GCP Vertex AI**, extracting structured insights from 1,000+ job descriptions. Built personal RAG projects for chatting with custom data using **Google Gemini, Pinecone, and ChromaDB**. Currently building robust REST APIs with **Fastify, Drizzle ORM, and PostgreSQL** at Ziply.ai.

WORK EXPERIENCE

Backend Developer – Ziply.ai, India

Apr 20, 2026 – Present

Dental Practice Management System

- Designing and developing RESTful APIs using **Fastify** and **TypeScript** for a dental practice management platform
- Modelling and managing relational data with **PostgreSQL** and **Drizzle ORM**, ensuring schema integrity and query performance
- Using **pgAdmin** for database administration, query profiling, and performance tuning
- Architecting cloud-native backend services on **AWS** following industry best practices for scalability and reliability

Tech Stack: Node.js, TypeScript, Fastify, Drizzle ORM, PostgreSQL, pgAdmin, AWS

Node.js Backend Developer – Worko.ai, India

Jul 2024 – Mar 2026

worko.ai

AI / ML Engineering

- Pulled **1,000+ job descriptions daily** from the Bright Data platform via automated pipelines and stored raw data in AWS S3 for processing
- Designed and executed **AWS Bedrock batch inference** jobs on S3-stored job descriptions to extract structured data — skills, experience requirements, and qualifications — at scale
- Built **GCP Vertex AI batch inference** pipelines using Google Cloud Storage (GCS) as the data source, generating high-quality text embeddings for semantic understanding
- Implemented **semantic job search** using **GCP Firestore Vector Database**, enabling similarity-based matching between candidate profiles and job descriptions
- Implemented **AI-powered job matching** combining vector search results with business rules to surface the most relevant opportunities for candidates

Backend & Infrastructure

- Designed and developed **serverless microservices** using AWS Lambda & Serverless Framework, reducing infrastructure overhead by ~40%
- Built advanced job search using **OpenSearch** with full-text search and multi-filter capabilities across 10,000+ listings
- Developed **real-time chat system** via AWS WebSocket API, enabling instant candidate–recruiter communication
- Designed event-driven architecture with EventBridge, SQS, and SNS for decoupled, fault-tolerant workflows
- Integrated WhatsApp API for multi-channel communication; implemented secure **RBAC authentication** using AWS Cognito
- Optimised DynamoDB queries with Dynamoose; built automated job aggregation pipelines via cron jobs, eliminating 100% of manual data-entry
- Developed file upload system using AWS S3 & Google Cloud Storage; set up monitoring with AWS CloudWatch and CI/CD via CodeCatalyst
- Integrated Razorpay payment library for in-app recharge and subscription management

Tech Stack: Node.js, TypeScript, AWS (Lambda, S3, DynamoDB, API Gateway, Cognito, SNS, SQS, EventBridge, Bedrock, CloudWatch, CodeCatalyst, SAM), OpenSearch, Serverless Framework, GCP (Vertex AI, Firestore Vector DB, GCS), Bright Data, Razorpay

MERN Stack Developer Intern – 21 Spheres (Neobrim)

Jan 2024 – May 2024

- Built full-stack e-commerce application using **React.js + Redux** with end-to-end cart and order flow
- Developed RESTful backend APIs using **Node.js, Express.js, and MongoDB** (Compass + Atlas) with Mongoose ODM
- Implemented CI/CD pipeline using GitHub Actions, cutting manual deployment steps to zero
- Followed modular architecture and clean coding practices, improving team code-review efficiency

PROJECTS

RAG Chat With Your Data — Personal Project

- Built a **Retrieval-Augmented Generation (RAG)** pipeline in Node.js/TypeScript enabling natural-language chat over custom datasets
- Implemented two interchangeable vector-store backends: **Pinecone** (managed cloud vector database) for scalable production-style search, and **ChromaDB** (local vector database) for offline development and testing
- Used **Google Gemini** for generating text embeddings and producing grounded chat responses from retrieved context
- Built ingestion/setup scripts to chunk, embed, and upsert documents into both vector stores, with ULID-based document IDs for traceability

Tech Stack: Node.js, TypeScript, Pinecone, ChromaDB, Google Gemini API, RAG, Vector Search, Embeddings

GitHub: github.com/Rohith-Patel13/RAG-chat-with-your-data

GenAI Node.js with Gemini — Personal Project

- Explored and implemented multiple **Generative AI** capabilities in Node.js/TypeScript using the **Google Gemini API**
- Built **text embedding generation** and cosine-similarity search over embedded data, plus a **ChromaDB** vector-store integration for semantic retrieval
- Implemented **function calling / tool use** with Gemini for an agentic flight-reservation workflow
- Built audio-to-text translation, speech-to-text, text-to-speech, and text-to-image generation features using Gemini's multimodal APIs

Tech Stack: Node.js, TypeScript, Google Gemini API, ChromaDB, Vector Search, Embeddings, Function Calling, Speech-to-Text, Text-to-Speech, Text-to-Image

GitHub: github.com/Rohith-Patel13/GenAi-Nodejs-Gemini (branch: feature/rohith-gemini-gen-ai)

TECHNICAL SKILLS

Backend:	Node.js, TypeScript, Express.js, Fastify, Microservices Architecture, REST APIs, WebSockets
AI / ML:	AWS Bedrock (Batch Inference), GCP Vertex AI (Batch Inference), GCP Firestore Vector DB (Semantic Search), Bright Data, Embeddings, Retrieval-Augmented Generation (RAG), Google Gemini API, Vector Search, Cosine Similarity, Function Calling / Tool Use
Cloud:	AWS (Lambda, S3, DynamoDB, Cognito, SNS, SQS, EventBridge, Bedrock, API Gateway, CloudWatch, CodeCatalyst, SAM), GCP (Vertex AI, Firestore, GCS)
Databases:	PostgreSQL, DynamoDB, MongoDB, Firestore (Vector), AWS OpenSearch / Elasticsearch, SQL, Pinecone (Vector DB), ChromaDB (Vector DB)
ORM / ODM:	Drizzle ORM, Dynamoose, Mongoose
Tools:	GitHub, pgAdmin, AWS SAM, Serverless Framework, Webpack, Razorpay
Soft Skills:	Problem-Solving, System Design, Communication, Cross-functional Collaboration

EDUCATION

B.Tech in Computer Science	2020 – 2024
CMR Technical Campus, Hyderabad	